

Roshan Srinivasan

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LinkedIn | Github | Portfolio



PROFILE

MSc Data Science student at TUHH, comfortable across the full chain from raw data to reliable reporting and deployed models. At Fraunhofer ITMP I consolidated measurement data from several sources — inconsistent protocols, units and detection limits — validated and analysed it, and built the ML models on top. I turn data into clear reporting, building data models and dashboards that support decisions. Python and SQL end-to-end, AWS Cloud certified, with hands-on GenAI experience and a practical sense of when it's the right tool.

EXPERIENCE

Fraunhofer Institute for Translational Medicine and Pharmacology

05/2026 – 08/2026

Student Assistant

Hamburg, Germany

- COMBINE (EU IMI-funded research project): replaced manual data consolidation with an automated Python pipeline that unifies inconsistent schemas and measurement units across source files, producing a single 13,800-row dataset ready for ML modelling
- Benchmarked gradient boosting, neural network and graph-based models on the consolidated dataset, reaching ~ 0.87 AUC and matching the published baseline
- Handled real experimental artefacts including censored values below detection limits, replicate measurements and protocol differences between contributing labs
- Built a Streamlit dashboard for hospital medical data covering distributions, correlations and cohort comparisons, enabling doctors across clinics to explore the data without scientific support

Walwin (Project under TUHH “Fishing for Experience”)

11/2025 – 03/2026

Software Developer (Data Science)

Hamburg, Germany

- Conducted exploratory analysis on Excel-sourced scholarship data to assess accuracy and consistency ahead of downstream modelling
- Developed a dashboard used by students across countries, reducing data query time by 60%
- Coordinated sprint planning and team workflow as Scrum Master using Agile and Kanban methods

Motiv AI LLP

12/2024 – 04/2025

Data Science and Analytics Intern

Coimbatore, India

- Automated accuracy validation across multi-source data pipelines, raising the proportion of records usable for analysis from 25% to 40%
- Implemented completeness and consistency checks feeding downstream reporting and analysis

PROJECTS

Financial Data Quality & Revenue Reporting | Python, Excel, Power Query, Power BI (DAX) 02/2026 – 03/2026

- Automated the collection of financial data with a Python web scraping pipeline, producing structured datasets ready for analysis
- Defined and applied pandas validation rules to review Excel-based financial data for correctness, accuracy and validity, ensuring reliable inputs for reporting

- Prepared and transformed data with Power Query, designed the data model and built Power BI dashboards with DAX measures, turning revenue data into clear insights for decision-making

Local Multimodal RAG Research Assistant | *Python, LangChain, ChromaDB, Ollama* | [GitHub](#) **07/2026 – 08/2026**

- Built a fully local, privacy-preserving RAG system for querying research documents, combining BM25 and vector retrieval to return source-cited answers with no data leaving the machine
- Implemented a multimodal ingestion pipeline handling text, tables and diagrams via Docling and a local vision-language model, with a folder watcher for automatic indexing of new documents
- Served the pipeline through a FastAPI backend and React frontend, later extending it into a compliance document review tool

Data Consolidation & Quality Pipeline | *VBA, Excel, Python*

07/2026

- Built a VBA tool that scans a folder of inconsistently formatted Excel exports, auto-detects header rows, maps varying column names to a standard schema, and normalises units into a single consolidated table
- Added row-level quality flagging (PASS/WARN/REJECT) with reasons written to an audit log, keeping rejected records visible rather than dropping them
- Tested against real-world defects including offset headers, multi-sheet workbooks, German decimal commas, censored values and missing identifiers

EDUCATION

Technical University of Hamburg (TUHH)

M.Sc. Data Science

10/2025 – Present

Hamburg, Germany

- Coursework: Big Data, Data Analysis, Statistical Methods and Testing, Machine Learning, Natural Language Processing, Causal Data Science

Amrita University

B.Tech. Computer Science Engineering — 8.44 CGPA, First Class with Distinction

10/2021 – 05/2025

Chennai, India

SKILLS

Programming & Scripting: Python, SQL, C++, JavaScript/TypeScript

Analytics: Power BI (DAX), Power Query, Excel (VBA), Python (pandas, NumPy), Data Validation & Quality Checks

GenAI: LangChain, RAG pipelines, ChromaDB, Ollama, embeddings & vector search, multimodal ingestion (Docling), prompt engineering

Agentic AI: Tool-calling agents, external API integration, local-first LLM deployment

Tools & Platforms: Git/GitHub, Docker, Streamlit, n8n, FAST APIs, Node.js, Kafka, VS Code, Microsoft 365

Cloud: Amazon Web Services (Certified Cloud Practitioner)

Project Management: Agile/Scrum, Kanban, sprint planning, stakeholder communication

CERTIFICATES

AWS Certified Cloud Practitioner

Databases and SQL for Data Science with Python — IBM

Introduction to Big Data with Spark and Hadoop — IBM

PUBLICATIONS

IEEE Xplore: “Optimizing Lung Cancer Classification Prediction Based on Random Forest and SMOTE

Techniques” — Random Forest model on a lung cancer dataset reaching 98% accuracy and 99% F1 score, using SMOTE to address class imbalance